

Begin

Sensors Data
Acquisition

$[u_k, g_B, m_B]$

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k$$

$$P_k^- = AP_{k-1}A^T + Q$$

$$\hat{y}_k^- = C\hat{x}_k^-$$

Predict

$\hat{x}_k, P_k$

$\hat{x}_k^-, P_k^-, \hat{y}_k^-$

$$K_k = \frac{(P_k^- C^T)}{(CP_k^- C^T + R)}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - \hat{y}_k^-)$$

$$P_k = (I - K_k C)P_k^-$$

Update